

NOWRA, Australia

WMO: 947500

Lat: 34.9469S

Lon: 150.5353E

Elev: 105

StdP: 100.07

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.3	5.2	-2.7	3.1	16.5	-1.2	3.5	14.1	13.5	12.8	12.4	13.0	2.9	290	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	33.4	20.1	30.8	20.1	28.8	19.8	23.1	28.3	22.2	26.9	21.5	25.9	6.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.6	24.5	20.9	15.8	23.7	20.2	15.0	23.1	69.0	28.4	65.7	27.1	63.1	26.0	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.8	8.5	DB	1.9	39.4	1.0	2.5	1.1	41.3	0.5	42.7	-0.1	44.2	-0.8	46.0	(4)
(5)			WB	1.1	24.7	1.1	1.1	0.4	25.5	-0.3	26.1	-0.9	26.7	-1.6	27.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	21.4	21.1	19.8	17.1	14.4	12.2	11.4	12.2	14.5	16.4	18.1	20.1	(6)	
	DBStd	4.34	3.00	2.69	2.43	2.37	2.13	1.75	1.83	2.25	2.81	3.17	3.31	2.99	(7)	
	HDD10.0	14	0	0	0	0	1	2	7	4	1	0	0	0	(8)	
	HDD18.3	1039	5	3	12	50	123	185	216	189	121	77	43	14	(9)	
	CDD10.0	2403	354	311	305	214	137	68	49	74	134	200	244	312	(10)	
	CDD18.3	387	101	82	58	14	1	0	0	1	5	19	37	68	(11)	
	CDH23.3	2973	747	527	323	101	9	0	1	7	85	244	355	575	(12)	
(13)	CDH26.7	1034	292	179	86	21	0	0	0	1	20	86	134	216	(13)	
(14)	Wind	WSAvg	3.9	3.5	3.4	3.3	3.3	3.9	4.2	4.5	4.6	4.2	3.9	3.7	3.7	(14)
(15) Precipitation	PrecAvg	1075	99	115	127	106	88	113	50	69	71	91	100	70	(15)	
	PrecMax	1929	333	450	429	503	328	609	154	365	235	412	585	226	(16)	
	PrecMin	463	10	2	2	1	2	8	0	1	5	4	2	2	(17)	
	PrecStd	366	70	109	118	111	77	120	43	89	60	83	104	55	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	35.5	31.8	29.5	24.7	20.9	21.0	24.2	29.2	33.1	35.1	36.2	(19)	
		MCWB	21.1	20.8	20.0	16.7	15.5	13.5	12.2	14.1	15.3	16.8	18.1	20.0	(20)	
	2%	DB	32.2	31.2	28.9	25.9	22.4	18.7	18.6	21.2	25.7	28.9	30.3	31.6	(21)	
		MCWB	21.4	22.3	20.6	17.4	15.0	12.8	11.6	12.6	14.4	16.4	18.8	19.1	(22)	
	5%	DB	29.8	28.5	26.7	23.8	20.6	17.4	17.0	19.2	23.1	25.9	27.0	28.6	(23)	
		MCWB	21.5	21.3	20.0	16.9	14.4	12.1	10.9	11.7	13.9	16.0	18.7	19.5	(24)	
	10%	DB	27.3	26.4	24.7	22.0	19.0	16.1	15.7	17.4	20.6	23.2	24.6	26.4	(25)	
		MCWB	20.6	20.6	19.0	16.4	13.8	11.4	10.4	11.1	13.4	15.5	17.7	19.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.1	24.4	23.3	20.2	18.3	15.6	14.5	15.8	18.0	20.0	22.1	23.0	(27)	
		MCDB	30.6	29.8	27.4	24.9	20.7	18.3	17.5	20.3	23.2	25.7	28.4	29.1	(28)	
	2%	WB	22.9	23.1	22.1	19.0	17.0	14.2	13.2	14.1	16.5	18.6	20.6	21.8	(29)	
		MCDB	28.5	28.1	25.6	23.0	19.1	16.5	15.8	18.1	21.7	24.0	26.6	27.4	(30)	
	5%	WB	22.0	22.2	21.1	18.1	15.8	13.3	12.2	13.0	15.5	17.6	19.5	20.9	(31)	
		MCDB	27.1	26.3	24.6	21.6	18.7	15.5	14.9	16.9	20.3	22.7	24.8	26.1	(32)	
	10%	WB	21.3	21.3	20.1	17.3	14.7	12.6	11.3	12.0	14.3	16.6	18.5	20.0	(33)	
		MCDB	25.8	24.9	23.2	20.6	18.0	14.9	14.3	15.8	18.8	21.4	23.0	24.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	8.9	9.0	9.5	9.2	8.3	9.0	10.2	11.0	11.1	10.3	10.2	(35)	
		MCDBR	15.1	13.6	13.1	12.6	11.6	10.2	11.2	13.1	15.6	16.7	16.0	15.6	(36)	
	5% WB	MCWBR	6.1	5.7	6.0	5.9	5.8	5.5	5.8	6.3	6.6	6.8	6.4	5.8	(37)	
		MCWBR	12.4	11.1	10.7	10.7	9.5	8.2	9.2	11.2	13.0	13.2	13.3	12.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.396	0.378	0.354	0.326	0.309	0.299	0.290	0.300	0.331	0.348	0.379	0.381	(40)	
		taud	2.422	2.503	2.549	2.603	2.598	2.629	2.642	2.604	2.515	2.494	2.441	2.452	(41)	
	Ebn at Noon		940	936	923	895	856	842	872	911	932	955	951	959	(42)	
		Edh at Noon	123	109	97	82	72	65	68	80	99	109	120	120	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.35	5.41	4.65	3.74	2.97	2.31	2.73	3.60	4.84	5.67	5.97	6.51	(44)	
	RadStd		0.52	0.48	0.37	0.32	0.17	0.20	0.22	0.21	0.30	0.46	0.51	0.52	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.43	+0.44	N/A	N/A	N/A	N/A	N/A	-114	+193	+93	(46)	
(47)	Regional (32 neighbors)	+0.44	N/A	N/A	+0.76	N/A	N/A	-1	-85	+182	+93	(47)	

Nomenclature: See separate page